

Lecture Syllabus

Module Code	Semester	ECTS	SWS	Lecture	Tutorial	Lab
MECH-M-DUAL-3-AMR	3	5	3	3	0	0
Course Name	Autonomous Mobile Robotics					
Lecturer	D. T. McGuiness, Ph.D (Daniel.McGuiness@mci.edu) (4A-434c)					
Study Programme	Mechatronik Automation Robotics AI					
Official Name	Autonome mobile Robotik				Lingo	English
Lecture Prerequisites	The student should be comfortable with working with either Python and/or C++ and should have an beginner level knowledge of the Linux environment and the command line interface					
Course Objectives	The goal of this lecture is to give you a better view of how mobile robotics works, the different challenges it faces and methods to overcome them. This lecture you will use ROS, a versatile software capable of doing pathfinding and localisation, a perfect fit for use in mobile robotics.					
Primary Course Content	Lecture Homepage on GitHub WebBook					
Secondary Course Content(s)	<i>A very informal journey through ROS 2</i> by M. M. Bassa , <i>A Concise Introduction to Robot Programming with ROS2</i> by F. M. Rico , <i>Programming Principles and Practice using C++</i> by B. Stroustrup , <i>Introduction to Autonomous Mobile Robots</i> by R. Siegwart, et. al ,					
Homework(s) and Project(s)	Personal Assignment A (50) Personal Assignment B (50)					
Assessment Criteria	Assignment Type			Effect	Count	
	Personal Assignment A			50	1	
	Personal Assignment B			50	1	

Lecture Structure

Order	Topic	Units	Self Study
1	Locomotion	4	8
	<i>Introduction · Legged Mobile Robots · Wheeled Mobile Robots</i>		
2	Mobile Robot Kinematics	4	8
	<i>Coordinate Systems</i>		
3	Perception	4	8
	<i>Introduction · Active Ranging · Vision Based Sensors · Feature Extraction</i>		
4	Mobile Robot Localisation	4	8
	<i>Introduction · The problems of Noise and Aliasing · Localisation v. Hard-Coded Navigation · Representing Belief · Representing Maps · Probabilistic Map-Based Localisation · Other Examples of Localisation Methods · Building Maps</i>		
5	Welcome to Linux	4	8
	<i>Learning the Linux Command Line · Installation · Docker</i>		
6	Command Line Fundamentals	4	8
	<i>Introduction · The Structure of Commands · Helpful Keyboard Shortcuts for the Terminal · When you need help with Commands · Additional Information</i>		
7	Working with Files and Folders	4	8
	<i>Introduction · A Detailed Look in ls Command · Creating and Removing Folders · Move, Copy and Delete Files and Folders · Role to Users and sudo · File Permissions · Hard and Symbolic Links · The Linux File System · Common Command-Line Tools and Tasks · Advanced Topics</i>		
8	Installation	1	8
	<i>ROS 2 Humble Hawksbill · Auto-Install Script</i>		
9	ROS Concepts	4	8
	<i>Introduction · Publisher and Subscriber Architecture · Nodes - The Building Blocks · The Discovery Process · Communication Between Nodes · Topics · Services · Actions · Parameters · Working with Command Line · Launch File · Intermediate Concepts · Configuring Quality of Service · Advanced Concepts</i>		
10	Command Line Tools	4	8
	<i>Setting the Environment · Turtles and Graphs · A Deeper Look into Nodes · Working with Topics · Working with Services · Working with Parameters · A Practical Look into Actions · Launching Nodes</i>		
11	Client Libraries	4	8

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Order	Topic	Units	Self Study
	<i>Getting Started with Colcon · Creating a Workspace · Creating a Package · Writing a Simple Publisher and Subscriber · Writing a Simple Service and Client · Creating Custom msg and srv Files · Using Parameters in a Class · Managing Dependencies · Creating an Action · Writing an Action Server and Client · Writing a Launch File</i>		
12	Transform Library	4	8
	<i>A Gentle Introduction · Writing a Static Broadcaster · Writing a Listener · Adding a Frame · Writing a Broadcaster</i>		
13	Sum	45	90

- Any major announcements will be made on SAKAI regarding any possible date/content/structural changes for the assignment(s), exam(s).
- Any lecture material will be posted at the lectures corresponding GitHub home-page. The link will be present on the lectures SAKAI homepage.
- If there are any questions regarding course content/exams/assignments please do not refrain from contacting me (Daniel.McGuiness@mci.edu).